

PDF 7: Servlets

1. A Servlet is a Java program that runs within a:

- a) Web browser
- b) Database server
- c) Servlet container/engine
- d) Client application

Answer: c

2. Servlets are a replacement for which older technology?

- a) JSP
- b) JDBC
- c) CGI programs
- d) HTML forms

Answer: c

3. How does a Servlet container handle multiple simultaneous requests?

- a) Forks a new process for each request
- b) Spawns a new thread for each request
- c) Queues all requests serially
- d) Creates a new Servlet instance for each request

Answer: b

4. What is the Servlet lifecycle order?

- a) init → service → destroy
- b) start → run → stop
- c) load → execute → unload
- d) create → process → kill

Answer: a

5. Which of the following manages the lifecycle of a Servlet?

- a) The developer

- b) The web browser
- c) The Servlet container
- d) The database

Answer: c

6. Which interface must a class extend to become a Servlet?

- a) GenericServlet
- b) HttpServlet
- c) ServletRequest
- d) WebServlet

Answer: b

7. Instance variables in a Servlet are:

- a) Thread-safe because each request has its own
- b) NOT thread-safe because multiple threads share the same Servlet object
- c) Automatically synchronized
- d) Destroyed after each request

Answer: b

8. Which scope stores attributes available to all servlets in a web application?

- a) Request scope
- b) Session scope
- c) Application (context) scope
- d) Page scope

Answer: c

9. What object is used to forward a request from a Servlet to a JSP?

- a) HttpSession
- b) RequestDispatcher
- c) ServletConfig
- d) HttpServletResponse

Answer: b

10. What is the difference between forward() and include() in RequestDispatcher?

- a) forward() keeps the original response; include() replaces it
- b) forward() cannot have any data written to response before calling; include() can
- c) Both do the same thing
- d) include() only works with HTML files

Answer: b

11. Which method retrieves a request parameter in a Servlet?

- a) request.getAttribute()
- b) request.getParam()
- c) request.getParameter()
- d) request.getValue()

Answer: c

12. Parameter values in Servlet requests are always of type:

- a) Object
- b) Integer
- c) String
- d) Byte

Answer: c

13. Attribute values in Servlet scopes are of type:

- a) String
- b) Integer
- c) Object
- d) Boolean

Answer: c

14. Which scope uses HttpSession to store attributes?

- a) Application scope
- b) Request scope
- c) Page scope
- d) Session scope

Answer: d

15. What does the JEE Web Container provide? (Select the best answer)

- a) Only HTTP support
- b) Manages lifecycle of Servlets/JSPs, session management, security, and resource management
- c) Only database connectivity
- d) Only static file serving

Answer: b

16. Which method in ServletContext gets an init parameter?

- a) getServletParam()
- b) getInitParameter(name)
- c) fetchParam(name)
- d) readConfig(name)

Answer: b

17. Local variables in a Servlet's service methods are:

- a) Shared across all threads
- b) Thread-unsafe
- c) Thread-safe
- d) Stored in the session

Answer: c

18. CGI creates a new _____ per request, while Servlets create a new _____:

- a) Thread; process
- b) Process; thread

- c) Object; session
- d) Session; object

Answer: b

19. What does `setAttribute(String name, Object object)` do in a Servlet?

- a) Gets a parameter from the request
- b) Stores an object in the specified scope with a given name
- c) Removes an attribute
- d) Sets a cookie

Answer: b

20. A Servlet can communicate with which of the following? (Best answer)

- a) Only the client browser
- b) Clients, the container, other server resources, and external resources like databases
- c) Only the database
- d) Only other Servlets

Answer: b

21. Request scope attributes are available:

- a) For the life of the application
- b) For the life of the user's session
- c) Until the `doGet` or `doPost` method completes
- d) Indefinitely until manually removed

Answer: c

22. Which object is used to get the `ServletContext` from a Servlet?

- a) `HttpSession`
- b) `HttpServletRequest`
- c) `getServletContext()` inherited from `GenericServlet`
- d) `RequestDispatcher`

Answer: c

23. The path supplied to `getRequestDispatcher()` starting with `/` is relative to:

- a) The servlet's location
- b) The current context root
- c) The server root
- d) The user's home directory

Answer: b

24. What advantage do Servlets have over CGI in terms of performance?

- a) Servlets are compiled each time; CGI is not
- b) Servlets are compiled and loaded only once; CGI compiles and loads for every request
- c) Servlets use more memory
- d) CGI uses threads, Servlets use processes

Answer: b

25. Which HTTP method in a Servlet handles GET requests?

- a) `doPost()`
- b) `doRequest()`
- c) `doGet()`
- d) `service()`

Answer: c

26. Session attributes may not be completely thread-safe because:

- a) Sessions are stored in the database
- b) Multiple simultaneous requests from the same session are possible
- c) Sessions expire too quickly
- d) The Servlet container deletes them randomly

Answer: b

27. Which of the following is a client of a Servlet?

- a) Only a web browser

- b) A browser, an applet, or a Java application
- c) Only Java applications
- d) Only mobile devices

Answer: b

28. What is `getServletConfig().getInitParameter(name)` used for?

- a) Getting request parameters
- b) Getting session attributes
- c) Getting initialization parameters from the Deployment Descriptor
- d) Setting context attributes

Answer: c

29. Context (application) scope attributes are inherently:

- a) Thread-safe
- b) Thread-unsafe (shared by all servlets)
- c) Scoped per request
- d) Destroyed on each request

Answer: b

30. What does `removeAttribute(String name)` do in `HttpSession`?

- a) Invalidates the session
- b) Deletes the named attribute from the session
- c) Removes the session cookie
- d) Clears all session attributes

Answer: b